Dividing Polynomials (2.4) Name ____________________

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write "2" in the blank. Solve that problem, find your answer and write "3" in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $x + 4$ # ______ Find the remainder when $f(x) = x^2 - 4$ is divided by $(x - 3)$.	ANSWER: 17 # ______ List the possible rational zeros of $f(x) = x^2 - 2$.
ANSWER: 0 # ______ Find the remainder when $f(x) = x^3 - 2x + 1$ is divided by $(x + 1)$.	ANSWER: yes # ______ Divide $x^2 - 9$ by $(x - 3)$; give the quotient.

ANSWER: 5

________ By the Factor Theorem, $(x-k)$ is a factor of f exactly when $f(k)$ equals what?

ANSWER: 1

________ Is $(x-2)$ a factor of $f(x) = x^3 - 8$?

ANSWER: $2x + 5$

________ For $f(x) = 2x^3 - x + 3$, find $f(2)$.

ANSWER: $x + 3$

________ Divide $x^2 + 7x + 12$ by $(x+3)$; give the quotient.

ANSWER: 2

________ Divide $2x^2 + 3x - 5$ by $(x-1)$; give the quotient.

ANSWER: $\pm 1,\ \pm 2$

________ Find the remainder when x^4 is divided by $(x-1)$.